

Radical Expressions and Rational Exponents — Finding the Illegal Split Answer Key

Algebra 2

Quick Answers

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|---------|------------------------|-------|
| 1. 5 | 4. $8\sqrt{2}$, 11.31 | 7. 29 |
| 2. 10 | 5. 13 | 8. 0 |
| 3. 3.46 | 6. 16, 16 | |
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Curriculum

Level: Algebra 2

HSA-REI.A.2 – *problems 6, 8*

HSN-RN.A.1, HSN-RN.A.2 – *problems 1, 2, 3, 4, 5, 7, 8*

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

1. If they got 7: took the root of each term and added, instead of adding first and then taking one root
 2. If they got 14: distributed the radical across the sum
 3. If they got 4.88: replaced the root of the sum with the sum of the roots
 4. If they got 8.25: merged two separate radicals into one radical of the sum
 5. If they got 17: gave each term its own one-half exponent, which is the same illegal split written another way
 6. If they got 4: split the radical into root x plus root nine, then squared what was left
 7. If they got 41: added the two legs instead of taking the root of the sum of the squares
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1. Compute $\sqrt{9+16}$, and compare it with $\sqrt{9} + \sqrt{16} = 7$.

Step 1. The radical sign is a grouping symbol, so everything under it is done first: $9+16 = 25$.

Step 2. Take one square root of that single number: $\sqrt{25} = 5$.

Step 3. Compare with the split version: $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. The two results differ by 2, so $\sqrt{a+b}$ and $\sqrt{a} + \sqrt{b}$ are not the same expression — one counterexample is enough to retire the rule for good. 5

Teaching point: the split is tempting because it looks like the distributive property. Distribution belongs to multiplication over addition; a radical is a power, and powers do not distribute over sums.

2. Priya: $\sqrt{36 + 64} = \sqrt{36} + \sqrt{64} = 14$.

Step 1. Locate the illegal step. It is the very first one: replacing $\sqrt{36 + 64}$ with $\sqrt{36} + \sqrt{64}$. Nothing after it can repair the answer, because the wrong quantity is already being evaluated.

Step 2. Add under the radical first: $36 + 64 = 100$.

Step 3. Take the root: $\sqrt{100} = 10$. 10

Step 4. Check the size of the error. Priya's 14 is 40% too large, and her numbers were chosen to be perfect squares, which is exactly what makes the split look safe.

3. Bo: $\sqrt{5 + 7}$ reported as 4.88.

Step 1. Bo replaced $\sqrt{5 + 7}$ with $\sqrt{5} + \sqrt{7}$ — the same split as in problem 2, hidden by decimals.

Step 2. Add inside first: $5 + 7 = 12$.

Step 3. $\sqrt{12} = 2\sqrt{3} = 3.4641\dots$, which rounds to 3.46. 3.46

Step 4. Compare: 3.46 against Bo's 4.88. When nothing comes out whole, the only defence is the rule itself, not the look of the answer.

4. Dana: $\sqrt{50} + \sqrt{18}$ written as $\sqrt{68} \approx 8.25$.

Step 1. Dana ran the split backwards: two radicals were merged into one radical of a sum. It is illegal in that direction too.

Step 2. Simplify each radical with the product rule, which IS legal because each radicand is a product: $\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$ and $\sqrt{18} = \sqrt{9 \cdot 2} = 3\sqrt{2}$.

Step 3. Now the two terms are like radicals — both are multiples of $\sqrt{2}$ — so they add like $5x + 3x$: $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.

Step 4. As a decimal, $8\sqrt{2} = 11.3137\dots \approx 11.31$, well clear of Dana's 8.25. $8\sqrt{2} \approx 11.31$

5. Kofi: $(25 + 144)^{1/2}$ evaluated as $25^{1/2} + 144^{1/2} = 17$.

Step 1. A rational exponent is a radical in other clothing: $(25 + 144)^{1/2} = \sqrt{25 + 144}$. Kofi's step is therefore the same split, and $(u + v)^{1/2} \neq u^{1/2} + v^{1/2}$ for the same reason that $(u + v)^2 \neq u^2 + v^2$.

Step 2. Add inside the parentheses first: $25 + 144 = 169$.

Step 3. Apply the one-half power to that single number: $169^{1/2} = \sqrt{169} = 13$. 13

Step 4. Kofi's 17 is what $5 + 12$ gives. Note that 5, 12, 13 is a right triangle: the legs never add to the hypotenuse, which is the same fact in geometric form.

6. Jamal: from $\sqrt{x + 9} = 5$ to $\sqrt{x} + \sqrt{9} = 5$, then $x = 4$.

Step 1. Test Jamal's answer in the ORIGINAL equation: $\sqrt{4 + 9} = \sqrt{13} = 3.6055\dots$, which is not 5. So $x = 4$ is not a solution, and the fault is the split in his first line.

Step 2. Solve correctly. Square the whole left side at once — squaring undoes the radical only when the radical is alone on that side, which it already is: $(\sqrt{x + 9})^2 = 5^2$, so $x + 9 = 25$.

Step 3. Subtract: $x = 25 - 9 = 16$. $x = 16$

Step 4. Check in the original equation: $\sqrt{16 + 9} = \sqrt{25} = 5 \checkmark$. Radical equations always get this check, because squaring can introduce extraneous roots even when every step is legal.

7. Patio diagonal: $\sqrt{20^2 + 21^2}$ feet, with Sam's $20 + 21 = 41$.

Step 1. Sam's answer is the split again: $\sqrt{a^2 + b^2}$ became $a + b$. Reject it on size alone — the diagonal of a patio cannot equal the two sides walked end to end.

Step 2. Square each leg first: $20^2 = 400$ and $21^2 = 441$.

Step 3. Add under the radical: $400 + 441 = 841$.

Step 4. Take one root of the total: $\sqrt{841} = 29$, and 29 is exact, since $29^2 = 841$. 29 ft

Step 5. Sanity check: 29 is longer than either side and shorter than $20 + 21 = 41$, which is exactly where a triangle's third side must land.

8. (a) Why the split fails, and (b) the one value of x that makes $\sqrt{x + 25} = \sqrt{x} + 5$ true.

(a) *Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.*

Square the right-hand side:

$$(\sqrt{a} + \sqrt{b})^2 = a + 2\sqrt{a}\sqrt{b} + b = a + b + 2\sqrt{ab}.$$

Squaring the left-hand side $\sqrt{a + b}$ gives just $a + b$. The two agree only when the extra cross term $2\sqrt{ab}$ is zero — that is, only when $a = 0$ or $b = 0$. For every other pair the split leaves out a whole positive term, which is why $\sqrt{a} + \sqrt{b}$ is always the larger of the two.

What a full-credit answer must contain: the squared form $a + b + 2\sqrt{ab}$, and the observation that the cross term $2\sqrt{ab}$ is what the split throws away.

(b) Part (a) already predicts the answer: the two sides can agree only when the cross term vanishes. Here $b = 25$ is fixed and not zero, so the only candidate is $x = 0$.

Step 1. Square both sides: $x + 25 = x + 10\sqrt{x} + 25$.

Step 2. Subtract $x + 25$ from both sides: $10\sqrt{x} = 0$, so $\sqrt{x} = 0$.

Step 3. Therefore $x = 0$, and the check confirms it: $\sqrt{0 + 25} = 5$ and $\sqrt{0} + 5 = 5$ ✓. $x = 0$